Ramp metering: the general total-time derivative

2 October 2026 — explanation of the current accounting identity

Let θ be one parameter of the open-loop meter-rate schedule. Let $P(t;\theta)$ count *all* traffic still in the accounting system: mainline, ramp, post-meter storage and external waiting. Let $A(t)$ be fixed exogenous arrival rate and $E(t;\theta)$ total *final exit rate*. E is neither cumulative exits nor local ramp merge entry. Compare the same initial population and arrivals over a fixed common horizon T that clears all compared plans. Here J is total system time, not mean time.

1. Start with conservation and complete time accounting. Internal transfers cancel in the population total, so

$$\dot{P}(t;\theta) = A(t) - E(t;\theta), \qquad J(\theta) = \int_0^T P(t;\theta)\,\mathrm{d}t.$$

Because initial mass and exogenous arrivals do not depend on the meter setting,

$$\frac{\partial P(t;\theta)}{\partial \theta} = -\int_0^t \frac{\partial E(u;\theta)}{\partial \theta}\,\mathrm{d}u.$$

Counting external waiting is essential: metering changes internal release, not the exogenous arrival population A.

2. Differentiate the cost and reverse the integration order. An exit-rate change at time u changes remaining population at every later time from u to T, giving duration weight $T-u$:

$$\begin{aligned}\frac{\partial J}{\partial \theta} &= -\int_0^T \int_0^t \frac{\partial E(u;\theta)}{\partial \theta}\,\mathrm{d}u\,\mathrm{d}t \\ &= -\int_0^T \int_u^T \frac{\partial E(u;\theta)}{\partial \theta}\,\mathrm{d}t\,\mathrm{d}u = \boxed{-\int_0^T (T-u)\frac{\partial E(u;\theta)}{\partial \theta}\,\mathrm{d}u.}\end{aligned}$$

This is the continuous accounting formula on slide 13. For CTM timestep h, K steps and left-endpoint accounting, the exact discrete companion is

$$P^{k+1} = P^k + h(A^k - E^k), \qquad J_h = h\sum_{k=0}^{K-1} P^k, \qquad \frac{\partial J_h}{\partial \theta} = -h^2 \sum_{k=0}^{K-2} (K-1-k)\frac{\partial E^k}{\partial \theta}.$$

3. Obtain exit sensitivity by the chain rule through the traffic model. The accounting identity does not itself determine how exits respond. In the CTM, the meter schedule $r^k(\theta)$ feeds release/storage, lateral exchange, forward cell flows and merge allocation. Write the whole state update as $x^{k+1} = F(x^k, r^k)$ and final exit rate as $E^k = H(x^k, r^k)$. With $z^k = \partial x^k/\partial\theta$ and fixed initial state,

$$z^0 = 0, \qquad z^{k+1} = F_x z^k + F_r \frac{\partial r^k}{\partial \theta}, \qquad \frac{\partial E^k}{\partial \theta} = H_x z^k + H_r \frac{\partial r^k}{\partial \theta}.$$

These sensitivities go into the weighted-exit formula. Moving final departures earlier reduces total time. Moving waiting between ramp and mainline while leaving total final exits unchanged gives zero total-time change.

Limits. Derivatives above assume an integrable sensitivity and fixed smooth active branches; min/max ties require separate event treatment. A blocked rate change can have zero sensitivity. Mean time divides both cost and derivative by fixed total mass. This is an analytical explanation, not numerical gradient verification or independent usefulness evidence.

Sources: current RM9 separate-resource/observer audit, sensitivity section; RM8 task/gradient screen. No model or review-gate decision is changed.